Even when our descendants are established on near-Earth asteroids and Mars and the moons of the outer Solar System and the Kuiper Comet Belt, it still won't be entirely safe. In the long run, the Sun may generate stupendous X-ray and ultraviolet outbursts; the Solar System will enter one of the vast interstellar clouds lurking nearby and the planets will darken and cool; a shower of deadly comets will come roaring out of the Oort Cloud threatening civilizations on many adjacent worlds; we will recognize that a nearby star is about to become a supernova. In the *really* long run, the Sun—on its way to becoming a red giant star—will get bigger and brighter, the Earth will begin to lose its air and water to space, the soil will char, the oceans will evaporate and boil, the rocks will vaporize, and our planet may even be swallowed up into the interior of the Sun.

Far from being made for us, eventually the Solar System will become too dangerous for us. In the long run, putting all our eggs in a single stellar basket, no matter how reliable the Solar System has been lately, may be too risky. In the long run, as Tsiolkovsky and Goddard long ago recognized, we need to leave the Solar System.

If that's true for us, you might very well ask, why isn't it true for others? And if it *is* true for others, why aren't they here? There are many possible answers, including the contention that they *have* come here—although the evidence for that is pitifully slim. Or there may be no one else out there, because they destroy themselves, with almost no exceptions, before they achieve interstellar flight; or because in a galaxy of 400 billion suns ours is the first technical civilization.

A more likely explanation, I think, issues from the simple fact that space is vast and the stars are far apart. Even if there were civilizations much older and more advanced than we—expanding out from their home worlds, reworking new worlds, and then continuing onward to other stars—they would be unlikely, according to calculations performed by William I. Newman of UCLA and me, to be here. Yet. And because the speed of light is finite, the TV and radar news that a technical civilization has arisen on some planet of the Sun has not reached them. Yet.